MATH 460 Exam # 3

1a.

	Age	Sex	Survived	Stand. Age	Salary (K)
A	27	M	0	0	80
B	30	F	1	0.057	70
C	80	F	1	1	90
D	50	M	0	0.434	60
E	60	F	0	0.623	10
F	70	F	1	0.811	100
G	55	F		0.528	

$$\text{Standardized Age} = \frac{\text{Age} - 27}{80 - 27}$$

$AG = \sqrt{(0-0.528)^2 + 12} = 1.131$

$BG = \sqrt{(0.057-0.528)^2 + 0^2} = 0.471$

$CG = \sqrt{(1-0.528)^2 + 0^2} = 0.472$

$DG = \sqrt{(0.434-0.528)^2 + 1^2} = 1.004$

$EG = \sqrt{(0.623-0.528)^2 + 0^2} = 0.095$

$FG = \sqrt{(0.811-0.528)^2 + 0^2} = 0.283$

Nearest neighor is E (Survived = 0)

1NN for G => Survived = 0

For 3NN, choose majority vote from the 3 nearest neighors:

E => Survived = 0

F => Survived = 1

B => Survived = 1

3NN for G => Survived = 1

The prediction does change if 3NN is used.

1b. The distances do not change, so the nearest neighbors are still the same.

1NN for G => Salary = $10,000 since salary for E is $10,000.

3NN:

E => 10,000

F => 100,000

B => 70,000

3NN = average of 3 nearest neighbors = $\frac{10,000 + 100,000 + 70,000}{3} = \$60,000$

2a. Uniform Weights

3 Nearest neighbors: B^0, A^1 and C^1

weight = 1

Predicted Prob. of E: $\frac{0+1+1}{3} = \frac{2}{3} > \frac{1}{2}$

3NN with uniform weights predicts 1 (x)

2b. Distance Weights

3 Nearest Neighbor: B^0, A^1 and C^1

weight = 1/distance

Predicted Prob. of E: $\frac{\frac{1}{1.4}(0) + \frac{1}{3}(1) + \frac{1}{3}(1)}{\frac{1}{1.4} + \frac{1}{3} + \frac{1}{3}} = \frac{0.667}{1.381} = 0.483$

$0.483 < \frac{1}{2}$

3NN with distance weights predicts 0 (o)

2c. Distance Weights

4 Nearest Neighbors: B^0, A^1, C^1 and D^1

weight = 1/distance

Predicted Prob. of E: $\frac{\frac{1}{1.4}(0)+\frac{1}{3}(1)+\frac{1}{3}(1)+\frac{1}{4}(1)}{\frac{1}{1.4}+\frac{1}{3}+\frac{1}{3}+\frac{1}{4}} = \frac{0.917}{1.631} = 0.562$

$0.562 > \frac{1}{2}$

4NN with distance weights predicts 1 (x)

3a.

	Item 1	Item 2	Item 3	Item 4	Item 5
Alice	5	3	3	4	
User 1	3	1	2	3	3
User 2	2	3	4	3	5
User 3	3	3	1	4	4
User 4	1	5	5	4	2

User-based kNN, k = 2, Manhantan distance

$d_{A1} = |5-3| + |3-1| + |3-2| + |4-3| = 6$

$d_{A2} = |5-2| + |3-3| + |3-4| + |4-3| = 5$

$d_{A3} = |5-3| + |3-3| + |3-1| + |4-4| = 4$

$d_{A4} = |5-1| + |3-5| + |3-5| + |4-4| = 8$

2 Nearest Neighbors: User 3 and User 2

Prediction = $\frac{4+5}{2} = 4.5$ (average of User 3 and User 2)

Since $4.5 \geq 4$, we recommend the item to Alice.

36. Item-based KNN, $k = 3$, cosine similarity

$$S_{51} = \frac{3(3) + 5(2) + 4(3) + 2(1)}{\sqrt{3^2 + 5^2 + 4^2 + 2^2} \times \sqrt{3^2 + 2^2 + 3^2 + 1^2}} = 0.936$$

$$S_{52} = \frac{3(1) + 5(3) + 4(3) + 2(5)}{\sqrt{3^2 + 5^2 + 4^2 + 2^2} \times \sqrt{1^2 + 3^2 + 3^2 + 5^2}} = 0.821$$

$$S_{53} = \frac{3(2) + 5(4) + 4(1) + 2(5)}{\sqrt{3^2 + 5^2 + 4^2 + 2^2} \times \sqrt{2^2 + 4^2 + 1^2 + 5^2}} = 0.803$$

$$S_{54} = \frac{3(3) + 5(3) + 4(4) + 2(4)}{\sqrt{3^2 + 5^2 + 4^2 + 2^2} \times \sqrt{3^2 + 3^2 + 4^2 + 4^2}} = 0.924$$

3 Nearest Neighbors: Item 1, Item 4, and Item 2

$$\text{Prediction} = \frac{5 + 4 + 3}{3} = 4 \quad \text{(Average of Items 1, 4 and 2)}$$

Since $4 \geq 4$, we recommend this item to Alice.